BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 092 · Lesson 092 · The Three Buses: City Highways

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What did PC advance to after sequential instruction at20?

Q2. What is MDR used for?

Q3. Which unit emits control signals?

Q4. Convert binary 10110000_2 to hexadecimal.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

address bus — Identifies location.

data bus — Transfers content bits.

control bus — Signals requests/status.

memory read — Retrieve contents of addressed location.

New learning

Observe — Address bus identifies the location the processor wants to access; typically the CPU drives address lines on a simple memory read.

Explain — Data bus transfers the contents between memory and processor; for ordinary memory reads the bits travel memory-to-CPU.

Evaluate — Control bus signals operation and coordination, including read/write; distinguish purpose and direction rather than treating all buses as one road.

Worked example

CPU reads memory[12]=99. Label each bus with what it carries.

1. Explain the mechanism — Address bus12; control bus READ; data bus returns99, placed in MDR. — Address12 and content99 are different types of information.

We Do · Practise with guidance

CPU writes value25 to memory address44. State bus role for each.

Explain the mechanism

Address bus44, data bus25 from CPU to memory, control bus WRITE.

On writing, data flows towards memory while address still identifies destination.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

A read from address7 returns value150. Identify address/data/control signals. [6]

Question 2

Explain role and typical direction of all three buses for a write to memory address20 of value9. [6]

Question 3

Evaluate why confusing addresses with contents can break CPU fetch models; use PC,MAR,MDR,CIR in example. [6]

Exit Ticket · Answer before checking

The address bus and data bus always carry the same binary number.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 21.

Q2. Data/instruction transfer with memory.

Q3. Control Unit.

Q4. $1011\ 0000_2 = B0_{16}$.

You Do pages · Independent questions

Q1. Address bus7 [2], control read [2], data bus150 [2].

Q2. Address CPU→memory20 [2], data CPU→memory9 [2], control WRITE [2].

Q3. PC points instruction address [1], MAR holds address [1], MDR carries instruction contents [2], CIR holds decoded/current instruction [2].

Exit Ticket · Model answer

False. Address selects location; data represents content.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: